Two-score settings with a continuous outcome

Claude, for the author (exploratory; 2026-10-28)

Common model and estimator

Each setting has two scores $Q = (Q_1, Q_2)$, covariates X, a continuous outcome Y, and a deployed rule $D = \mathbf{1}\{Q \in A_0\}$ that the data reproduce exactly. The model is

$$Q_j = \gamma_j^\top \tilde{X} + \eta_j, \quad j = 1, 2, \qquad X \perp \eta = (\eta_1, \eta_2), \qquad T_j = \gamma_j^\top \tilde{X},$$
$$Y = b(\eta) + X^\top \beta_1 + D\{\alpha(\eta) + X^\top \beta_2\} + \varepsilon, \qquad \mathbb{E}(\varepsilon \mid D, X, \eta) = 0.$$

For a rule A the target is the supported utility

$$U(A) = \mathbb{E}\big[I(\eta)\{\alpha(\eta) - c + X^\top \beta_2\}\mathbf{1}\{T + \eta \in A\}\big], \qquad I(\eta) = \mathbf{1}\{\epsilon \le \pi_0(\eta) \le 1 - \epsilon\}, \quad \pi_0(\eta) = P(T + \eta \in A_0),$$

the gain in Y per unit from treating the units the rule selects, counted only where the deployed rule gives overlap. Here $c = 0$ and $\epsilon = 0.05$. Rule classes: the weighted sum $\lambda Q_1 + (1 - \lambda)Q_2 \ge \phi$ with the deployed weight (only ϕ moves) or a chosen weight; a cutoff on each score with both required ($Q_1 \ge \phi_1$ and $Q_2 \ge \phi_2$); and with either enough.

Estimator (`experiments/methods/two_score.py`). Four folds with four roles (first stage, outcome, index, evaluation), rotated. OLS of each score on X; $\hat{\alpha}$ and $\hat{b}$ are tensor-product cubic splines in (η_1, η_2) with a second-difference penalty chosen by 5-fold cross-validation, and X enters linearly in each arm; $\hat{U}(A)$ is the pairwise average over $\hat{\eta}_i$ (evaluation fold) and $(\hat{T}_j, X_j)$ (index fold). Each rotation maximizes $\hat{U}$ over each class; the reported rule is the average of the four. *No bootstrap here*: the range over the four rotations is shown as a rough gauge of stability, not a confidence interval. In simulations the feasible estimator is root-n from about $n = 64$k, with a variance 20–40 times the oracle's.

Figures (one panel set per setting; averages over the four rotations): (a) $\hat{\alpha}(\eta)$, the effect at the covariate mean, drawn only where $\hat{U}$ counts (inside the trim, dashed) and where there are data, with the density of $\hat{\eta}$ in grey and $\hat{\alpha} = 0$ solid; (b) $\hat{U}(\phi)$ with the deployed weights; (c) $\hat{U}(\lambda, \phi)$; (d) $\hat{U}(\phi_1, \phi_2)$ for both cutoffs. $\hat{U}$ is in units of Y per unit of the evaluation fold.

Settings (one per page): Texas A/B; Florida B/C; North Carolina B/C; North Carolina low-performing designation; NYC Progress Reports A/B.

Left out. Binary outcomes (Chile enrollment, gainful-employment closure, bank failure). Settings where every rule class went to the edge of its range because the estimated effect is flat, null or too noisy: Wisconsin, California Dashboard, CMS hospital penalty, dialysis payment cuts, nursing-home staffing, FDIC capital (loan growth; banks also bunch at the cutoffs), census tracts, NYC C/D. Too small for four folds: Louisiana ($n = 1{,}089$), Indiana (2,590).

Common caveats. $X \perp \eta$ fails in every setting (the noise is heteroskedastic in T); schools repeat across years; and for schools, next year's score reuses some of this year's test data. The effect is identified only on the trimmed region, near the deployed boundary.

Code: `experiments/scripts/two_score_writeup.py` (figures, `numbers.json`); datasets under `experiments/datasets/`.

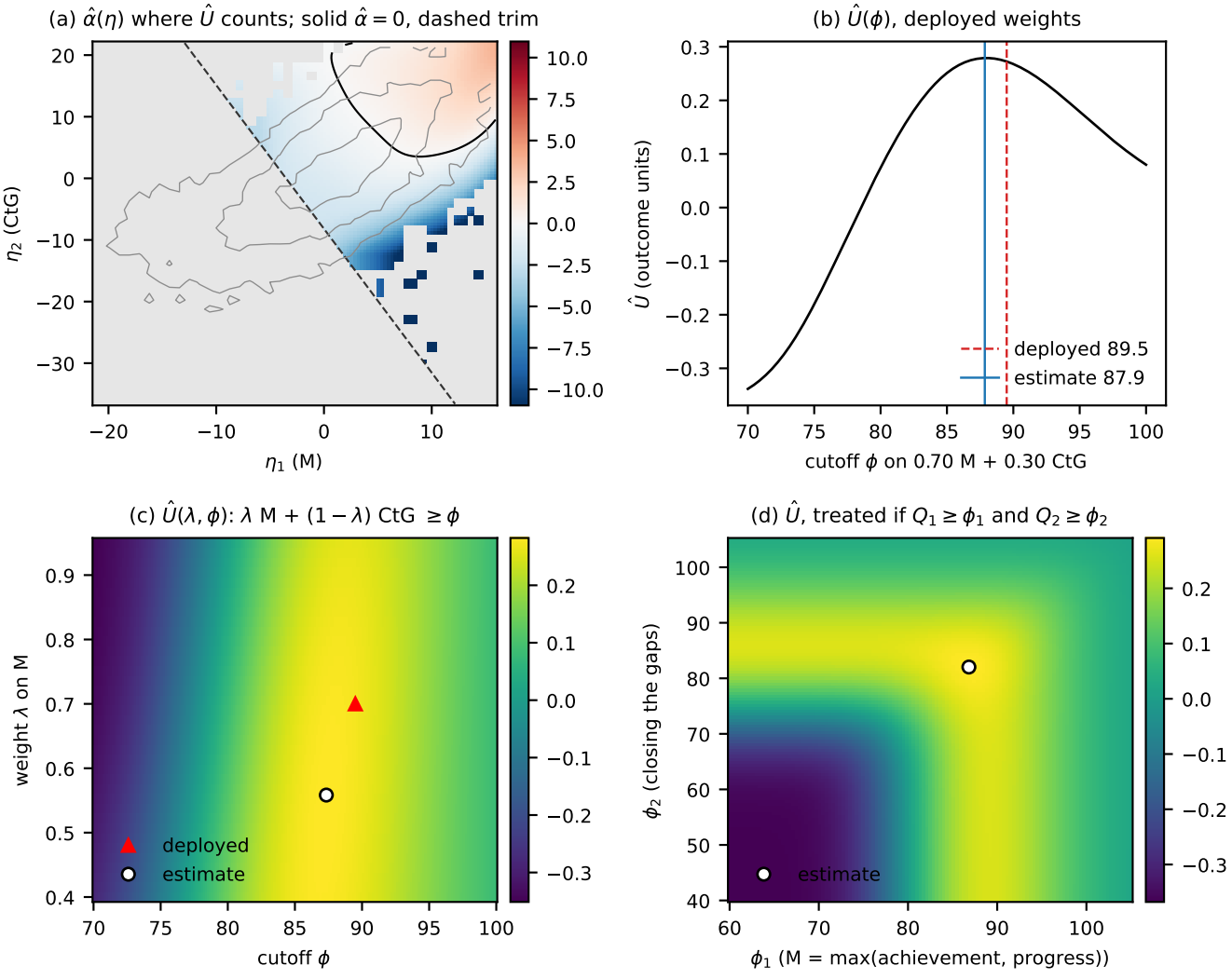
1. Texas A–F campus ratings: the A/B boundary

Data. TEA campus ratings, 2019 and 2022–2025; non-alternative campuses with all three domain scores, outside the four districts with local accountability; $n = 38{,}075$ campus-years, 22.5% rated A (`tx_school_ratings`).

Scores. $Q_1 = M = \max(\text{Student Achievement}, \text{School Progress})$ and $Q_2 =$ Closing the Gaps, each a scaled score 0–100. **Deployed rule.** A if $\text{round}(0.7M + 0.3\,\text{CtG}) \geq 90$, i.e. $0.7Q_1 + 0.3Q_2 \geq 89.5$ (reproduces every published score outside the local-accountability districts). X (34 columns): % economically disadvantaged, % emergent bilingual, log students, lowest and highest grade, charter, school type, district log enrollment and % disadvantaged, ESC region, year. Y: the next rated year's overall scaled score (sd 10.4).

Fit. R^2 of Q on X: 0.37, 0.30; $\text{corr}(\hat{\eta}_1, \hat{\eta}_2) = 0.73$; 66% of evaluation units inside the trim. In the trimmed region $\hat{\alpha}$ ranges from -29 to 3.9 (mean -1.3; positive for 37%); $\text{sd}(X^\top \hat{\beta}_2) = 2.3$.

class	deployed	estimate	4 rotations	$\hat{U}$ (deployed 0.272)
weights fixed (0.7, 0.3)	$\phi = 89.5$	$\phi = 87.9$	87.0–88.9	0.281
weights chosen	$(0.7, 89.5)$	$(0.56, 87.3)$	λ 0.40–0.75; ϕ 86.8–88.5	0.287
both cutoffs	–	$(86.8, 82.0)$	ϕ_1 85.7–88.0; ϕ_2 79.1–83.7	0.294
either cutoff	–	unstable	3 of 4 rotations at the edge	0.270



Reasonable? The rule is reproduced exactly and the sample is the largest of the school settings. $\hat{U}(\phi)$ has a clear interior peak 1.6 points below the deployed cutoff, and the both-cutoffs optimum is stable across rotations. But the gain is about 0.01–0.02 points of a score with sd 10.4, the weight is poorly determined ($\hat{U}$ is flat in λ, panel c), and the large negative $\hat{\alpha}$ sits in a sparse corner. The 2023 and 2024 ratings were published after the next year's tests.

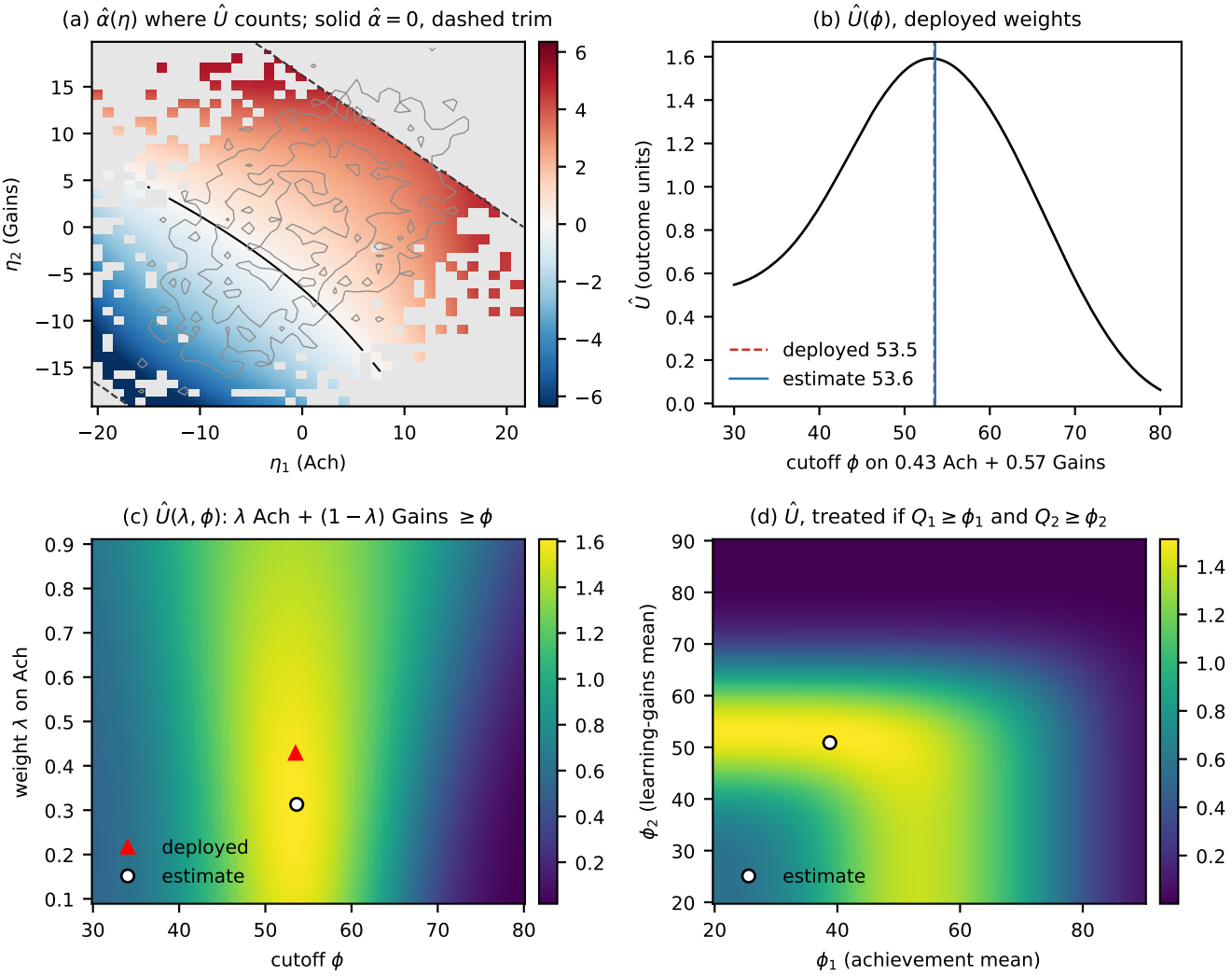
2. Florida school grades: the B/C boundary (elementary)

Data. FLDOE school grades, 2017–2019 and 2022; elementary schools graded on exactly seven components; $n = 6{,}940$ school-years, 56% graded B or better (`fl_school_grades`, `load_two_score`).

Scores. Q_1 = mean of ELA, math and science achievement (%), Q_2 = mean of the four learning-gains components (%). **Deployed rule.** B or better if the percent of points $(3Q_1 + 4Q_2)/7 \geq 53.5$ (published grades match for every school). X (73 columns): % economically disadvantaged, % minority, log enrollment, charter, Title I, collocated, district, year. Y: next year's percent of points (sd 13.3).

Fit. R^2: 0.72, 0.35; $\text{corr}(\hat{\eta}) = 0.53$; 89% inside the trim. Cross-validation picks the top of the penalty grid in every rotation, so $\hat{\alpha}$ is close to bilinear: from −8.8 to 5.2 (mean 0.3; positive for 59%); $\text{sd}(X^\top \hat{\beta}_2) = 5.7$.

class	deployed	estimate	4 rotations	$\hat{U}$ (deployed 1.592)
weights fixed (3/7, 4/7)	$\phi = 53.5$	$\phi = 53.6$	49.8–56.1	1.629
weights chosen	$(0.43, 53.5)$	$(0.31, 53.6)$	λ 0.22–0.43; ϕ 50.3–56.0	1.649
both cutoffs	–	$(38.8, 50.9)$	ϕ_1 20–55 (unstable); ϕ_2 48.1–53.8	1.570
either cutoff	–	$(61.4, 56.3)$	ϕ_1 56.7–65.7; ϕ_2 52.6–60.5	1.620



Reasonable? The estimated cutoff equals the deployed one, and the chosen weight puts more weight on learning gains (0.69 against 0.57). The rotations spread over 6 points, and the sample is small for four folds. With the penalty at the top of the grid the effect surface is essentially a plane, so the shape in panel (a) is imposed more than estimated. I have not checked whether the trim, which is centred on the deployed boundary, pulls the estimate toward it.

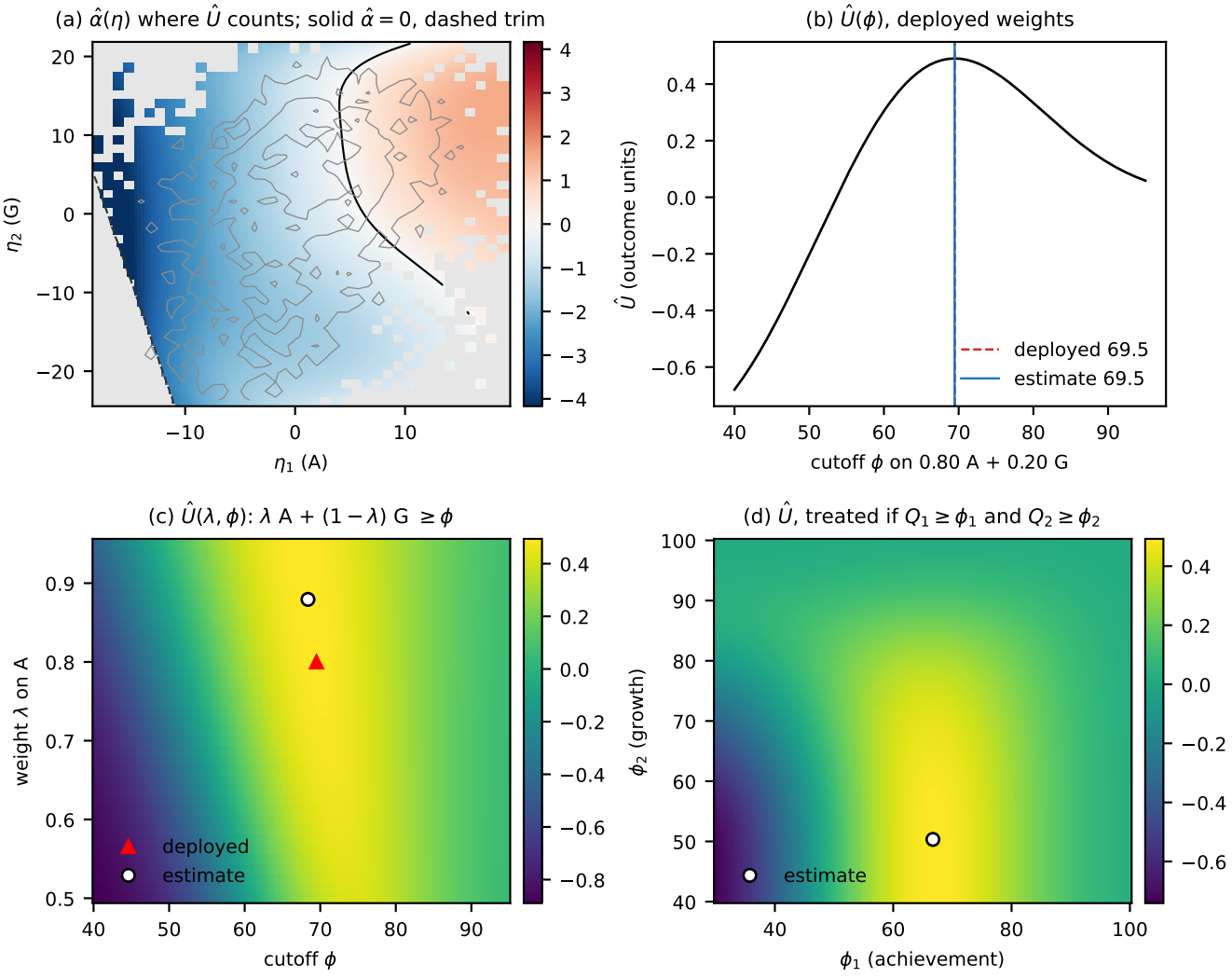
3. North Carolina school performance grades: the B/C boundary

Data. NC DPI school performance grades, 2018–2025; $n = 12{,}425$ school-years, 29.9% graded B or better (`nc_school_grades`).

Scores. Q_1 = achievement score, Q_2 = growth score (0–100). **Deployed rule.** B or better if $0.8Q_1 + 0.2Q_2 \geq 69.5$ (100% of published grades). X (129 columns): % economically disadvantaged, race shares, English learners, students with disabilities, log ADM and its change, % inexperienced teachers, charter, Title I, calendar, school type and special types, district, year. Y: next year's school performance score (sd 14.4).

Fit. R^2: 0.79 for achievement but 0.11 for growth; $\text{corr}(\hat{\eta}) = 0.41$; 96% inside the trim. $\hat{\alpha}$ from -4.9 to 1.6 (mean -1.0; positive for 26%); $\text{sd}(X^\top \hat{\beta}_2) = 4.4$.

class	deployed	estimate	4 rotations	$\hat{U}$ (deployed 0.490)
weights fixed (0.8, 0.2)	$\phi = 69.5$	$\phi = 69.5$	66.8–72.5	0.498
weights chosen	$(0.8, 69.5)$	$(0.88, 68.4)$	λ 0.75–0.95 (2 at edge); ϕ 64.3–73.1	0.505
both cutoffs	–	$(66.7, 50.3)$	ϕ_1 63.5–69.4; ϕ_2 39.9–68.1	0.502
either cutoff	–	achievement only	growth cutoff at the edge in all 4	0.486



Reasonable? The rule and sample are clean, but the estimate is the deployed cutoff with a 6-point rotation range, the gain is 0.01 points, and every class that can move the growth cutoff pushes it out of play. Growth is nearly unpredictable from X ($R^2 = 0.11$), so T has little spread in that direction and the trade-off between the scores is weakly identified.

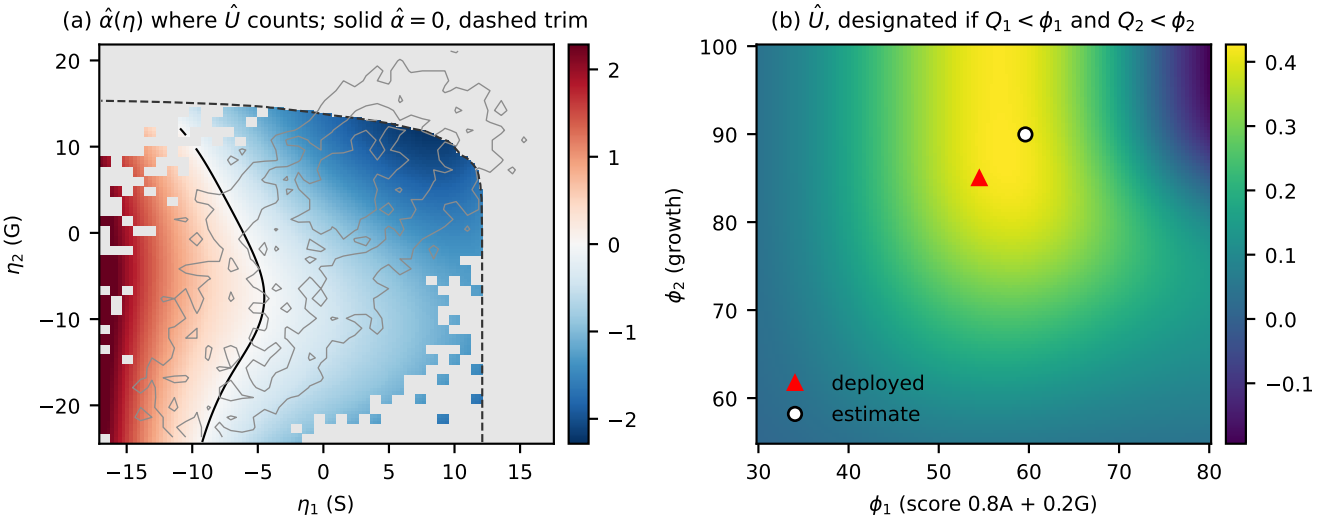
4. North Carolina low-performing designation (both cutoffs)

Data. As in setting 3; $n = 12{,}425$ school-years, 26.5% designated (`nc_school_grades`, `load_low_performing`).

Scores. $Q_1 = 0.8A + 0.2G$ (the school performance score), Q_2 = growth G. **Deployed rule.** Designated low-performing if $Q_1 < 54.5$ *and* $Q_2 < 85$ (a both-cutoffs rule on the low side; reproduces all 4,055 designations). Designation brings a required improvement plan and district and state support. X, Y: as in setting 3. $D = 1$ is designation.

Fit. R^2: 0.74, 0.11; $\text{corr}(\hat{\eta}) = 0.64$; 84% inside the trim. $\hat{\alpha}$ from -2.2 to 2.8 (mean -0.3; positive for 34%), positive for schools whose score is below what X predicts ($\eta_1 < -5$) and negative above; $\text{sd}(X^\top \hat{\beta}_2) = 4.6$.

class	deployed	estimate	4 rotations	$\hat{U}$ (deployed 0.413)
both cutoffs	$(54.5, 85)$	$Q_1 < 59.6$ and $Q_2 < 90.0$	ϕ_1 52.9–73.3; ϕ_2 86.2–99.2	0.494
either cutoff	–	unstable	3 of 4 rotations at the edge	0.489



Reasonable? This is the setting where $\hat{\alpha}$ changes sign inside the data in an interpretable way: designation helps the schools that score far below their predicted level and hurts the others. $\hat{U}$ has a clear ridge (panel b) and the estimate would designate more schools than the deployed rule. Three of the four rotations agree on $(53\text{–}58, 86\text{–}88)$; the fourth drifts to $(73, 99)$. The gain, 0.08 points, is the largest relative to Y among the five settings, but still small.

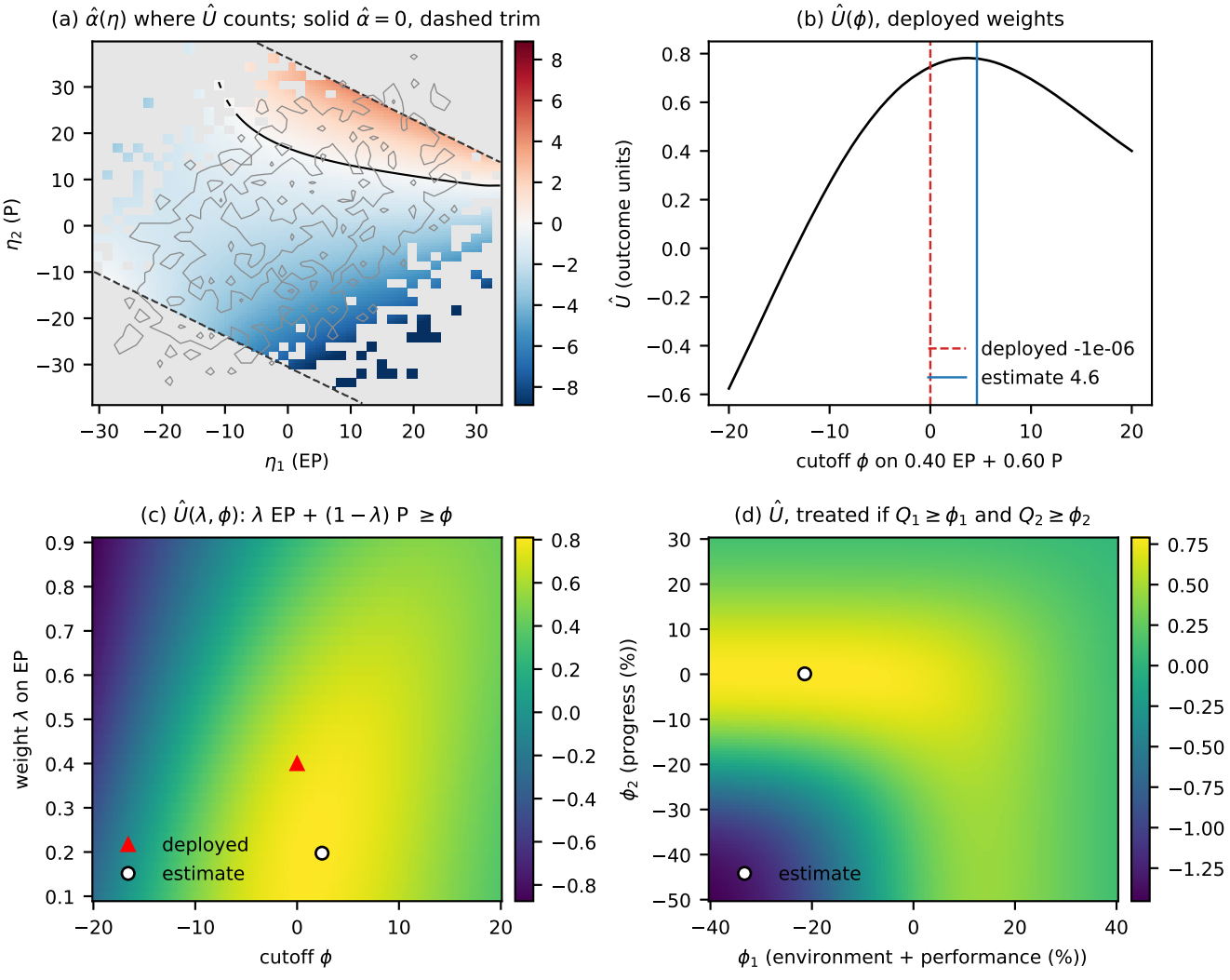
5. NYC Progress Reports: the A/B boundary

Data. NYC DOE Progress Reports for elementary, K-8 and middle schools, rating years spring 2008–2012; $n = 5{,}348$ school-years, 39.4% graded A (`nyc_progress_reports`, `load_two_score("AB")`).

Scores. With c the year's published cut for the school type, $Q_1 = 100(\text{Environment} + \text{Performance} + \text{credit} - 0.4c)/40$ and $Q_2 = 100(\text{Progress} - 0.6c)/60$: each in percent of its category maximum, net of its share of the cut. **Deployed rule.** A if $0.4Q_1 + 0.6Q_2 \geq 0$ (every published A/B grade). X (62 columns): poverty, ELL, special-education, race and male shares, log enrollment, tested share, the DOE peer indices, last year's percentile, school type, charter, community district, year (missing-value flags). Y: next year's overall score (sd 19.4).

Fit. R^2: 0.52, 0.31; $\text{corr}(\hat{\eta}) = 0.55$; 85% inside the trim. $\hat{\alpha}$ from -13.8 to 4.2 (mean -2.0; positive for 20%); $\text{sd}(X^\top \hat{\beta}_2) = 7.5$.

class	deployed	estimate	4 rotations	$\hat{U}$ (deployed 0.746)
weights fixed (0.4, 0.6)	$\phi = 0$	$\phi = 4.6$	-0.9–12.4	0.839
weights chosen	(0.4, 0)	(0.20, 2.4)	λ 0.10–0.30; ϕ -4.1–9.9	0.882
both cutoffs	–	$(-21.4, 0.1)$	ϕ_1 -38 to -4; ϕ_2 -6.7–10.1	0.871
either cutoff	–	unstable	ϕ_1 29–60	0.859



Reasonable? The direction is consistent across rotations: a higher A cutoff and more weight on progress (the chosen λ is 0.1–0.3 in every rotation). But the cutoff ranges over 13 points, n is small, and from 2009–10 the A share was fixed by design each year (the cut was set at a percentile), so a "different cutoff" also means a different A share. The category scores are percentile-based, and the metrics changed in 2009–10.